CSE 144 Final Project Report

Transfer Learning Challenge

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1 Introduction

The task is 100-class image classification on a small, heavily imbalanced dataset assembled from several source datasets. The training set has 1079 colored images of varying shapes spread across 100 classes (roughly 10 per class, but in practice 4 to 41 per class), and the test set has 1036 unlabeled images that must each be assigned one of the 100 class labels. The model is evaluated on Kaggle; the public leaderboard reports an estimated score on about 10% of the test set.

With only about 10 training images per class, training a large vision model from scratch is infeasible: it would overfit immediately and never see enough data to learn good features. Transfer learning sidesteps this. A backbone pretrained on hundreds of millions of images already encodes strong, general visual features; we only need to adapt a small number of parameters to the 100 target classes. The assignment explicitly requires loading strong pretrained weights and fine-tuning them.

We worked through a sequence of increasingly capable approaches (a frozen-feature linear probe, multi-view feature augmentation, a frozen cross-backbone ensemble, and a fine-tuned LoRA ensemble) and found that a single LoRA-fine-tuned SigLIP-2 backbone matched the best ensemble's leaderboard score at one third of the cost. The deployed model is one SigLIP-2 SO400M backbone at 378 px, LoRA-fine-tuned (rank 8 on the last 4 attention blocks) with an L2-normalized cosine head and a class-balanced loss. It reaches an out-of-fold (OOF) cross-validation accuracy of 0.9484 ± 0.0004 (3 seeds) and a public leaderboard score of 0.93636, well above the 0.60 baseline required by the assignment.

2 Dataset

The dataset has 100 classes and 1079 training images in total, imbalanced between 4 and 41 images per class (the assignment describes about 10 per class). There are 1036 test images, each of which is a scored row in the submission. We do not hold out a fixed validation split; instead we use 4-fold stratified cross-validation on the training set, which uses every training image for both training and validation across folds and yields an out-of-fold (OOF) accuracy that is a more reliable estimate than a single small held-out split.

The data follows the directory structure given in the assignment:

```
data/
  train/<class>/<n>.jpg   # 100 class folders, 1079 images
  test/<id>.jpg           # 1036 test images, all scored
  sample_submission.csv   # template for ID,Label format
```

Each training subfolder is named by its class string ("0" through "99"). Internally we build a class-name-to-index map for training, but predictions are written back out using the original class-folder name (Label equals the class string), so the submitted labels match the folders exactly regardless of the internal index order. This satisfies the assignment's requirement that class "0" maps to Label 0, class "1" to Label 1, and so on; a scrambled mapping would score no better than random. The submission is a two-column CSV with columns ID and Label, one row per test image, all 100 classes represented.

All images are resized to a fixed square input. Each backbone uses its own ImageNet-style normalization statistics (mean and standard deviation supplied by the timm model config). For the deployed SigLIP-2 model the input is 378 px. Training augmentation is kept light to limit overfitting on so few images: `RandomResizedCrop(378, scale=0.7-1.0, bicubic)` followed by a horizontal flip with probability 0.5, then tensor conversion and normalization. The evaluation and test transform is a deterministic "identity" resize to the model input size and center crop, then normalization; no augmentation is applied at test time (see the test-time augmentation ablation in Section 4).

3 Implementation

3.1 Model

The deployed backbone is SigLIP-2 SO400M (`vit_so400m_patch14_siglip_gap_378.v2_webli`) at 378 px, loaded from the timm model hub. We compared three strong pretrained backbones during the project: DINOv3 ViT-L/16 (self-supervised), SigLIP-2 SO400M (image-text contrastive), and AIMv2 ViT-L/14 (autoregressive image pretraining, distilled). After fine-tuning, SigLIP-2 was clearly the strongest single backbone on our cross-validation (OOF 0.9435 on the deploy seed, about 0.9401 mean across seeds, versus about 0.91 for DINOv3 and AIMv2). Its image-text contrastive pretraining transfers especially well to this varied, multi-source dataset, and it gains the most from fine-tuning, so we deploy SigLIP-2 alone.

SigLIP-2 is a global-average-pooling model with no CLS token, so we mean-pool its output tokens into a single 1152-dimensional feature vector. On top we attach a trainable cosine classifier head: features and per-class weight vectors are both L2-normalized, their dot product is scaled by a learnable scalar (initialized to 10), and the result is the 100-class logit. The cosine head ports the L2-normalized recipe that helped our earlier frozen probe into the fine-tuned head. A plain linear head remains the code default; the cosine head is opt-in.

We use LoRA (low-rank adaptation) rather than full fine-tuning. The pretrained backbone weights stay frozen; LoRA inserts small trainable low-rank matrices (rank $r = 8$, $\alpha = 16$, dropout 0.05) into the attention projections (`attn.qkv` and `attn.proj`) of the last 4 transformer blocks only. This keeps the trainable parameter count tiny, which is essential with about 1000 images: full fine-tuning of a 400M-parameter backbone would overfit immediately. Only the LoRA matrices and the head are trained; peak GPU memory stays well under the 16 GB budget.

3.2 Training

The loss is class-balanced cross-entropy with label smoothing 0.1. The class weights are the scikit-learn "balanced" weights computed per fold, $w_c = n_{\text{train}}/(n_{\text{classes}} \cdot \text{count}_c)$ with counts clamped to at least 1, which up-weight rare classes. This aligns the loss with the approximately class-balanced test set despite the 4-to-41 per-class training imbalance. The optimizer is AdamW with separate learning rates for the two parameter groups. Table 1 lists the hyperparameters. Each fold trains until validation accuracy stops improving for 5 epochs (it typically converges by epoch 10 to 15); the best-epoch validation softmax forms that fold's OOF predictions.

Hyperparameter	Value
LoRA rank r / α / dropout	8 / 16 / 0.05
Adapted modules	`attn.qkv` + `attn.proj`, last 4 blocks
Learning rate (LoRA / head)	10^{-4} / 10^{-3}
Weight decay	0.01
LR schedule	cosine decay, 10% linear warmup
Label smoothing	0.1
Precision	bf16 (autocast)
Batch size	16
Max epochs	30
Early stopping	validation accuracy, patience 5
Input resolution	378 px

Table 1: Training hyperparameters for the deployed model.

Training ran on a single NVIDIA RTX 5070 Ti (16 GB, Blackwell) with Python 3.14.0, PyTorch 2.9.0+cu128 (CUDA 12.8), timm 1.0.27, peft 0.19.1, scikit-learn 1.7.2, and numpy 2.3.2. The full set of training fine-tunes (3 seeds $\times$ 4 folds = 12) plus the deployed 4-fold ensemble runs comfortably on one GPU; peak memory is far below 16 GB. CPU works but is slow.

4 Experiments

The honest accuracy proxy throughout the project is 4-fold stratified-CV OOF accuracy on the 1079 training images. The public leaderboard is computed on only about 10% of the test set (roughly 110 images), so a one- or two-image swing there is within noise; the 1079-image OOF is the more reliable signal, and we trust it over the public leaderboard when the two disagree.

The baseline is a frozen DINOv3 ViT-L/16 feature extractor (CLS + mean-patch, 2048-d) feeding a per-fold standardized multinomial logistic-regression probe at 256 px, with identity + horizontal-flip test-time views. This reaches OOF 0.8851 and about 0.88 on the public leaderboard; it is our starting point and the reference for all later gains.

All tuning decisions are made on the shared 4-fold OOF, never on the public leaderboard (which we use only to confirm submission format and direction of transfer). For the frozen probe we cross-validate the regularization strength C with repeated stratified k-fold. For LoRA we fix the low-overfit-risk hyperparameters (rank, adapted blocks, resolution, augmentation) and sweep only choices with a clear accuracy or robustness rationale. We report results over 3 seeds (42, 43, 44) with mean and standard deviation.

The key experiments, in order:

1. *Resolution (frozen probe).* Going 256 $\to$ 384 $\to$ 512 px raised OOF from 0.8851 to 0.8934 to 0.8953. But the 512 px submission *regressed* on the public leaderboard (0.86363, down from about 0.88) even though CV went up. We attribute this to a train/test resolution shift (upsampling helped same-distribution CV folds but hurt the held-out test set) and reverted to 256 px. This episode set the project's rule: trust multi-seed OOF, and do not change test-time resolution.

2. *Test-time augmentation (TTA).* At 512 px, averaging in flipped/scaled views *hurt* OOF (identity alone 0.8953 vs. identity + hflip 0.8860). Multi-view ensembling of the frozen features ("multiview $K = 8$") was a separate, beneficial augmentation that lifted OOF to 0.9129 and the leaderboard to 0.90000, but view-TTA at inference consistently hurt.

3. *Resolution ensembling.* Averaging 256/384/512 models dragged the strong 512 signal down; the single best resolution always won.

4. *LoRA fine-tuning vs. frozen probe.* On a paired 4-fold OOF (same folds), LoRA-DINOv3 beat its frozen probe by +2.47 points (0.9132 mean vs. 0.8885), with the gap roughly 13× the seed spread – a decisive go. The LoRA submission scored 0.89090 (up from the baseline), confirming the CV gain transferred because test resolution was unchanged.

5. *Cross-backbone ensembles.* A frozen 3-backbone blend reached OOF 0.9314 and leaderboard 0.91818. Making every member a fine-tuned LoRA backbone lifted it to OOF 0.9404 (3-seed mean) and leaderboard 0.93636, and satisfied the assignment's fine-tuning requirement.

6. *Single backbone vs. ensemble (the deploy decision).* The fine-tuned ensemble's accuracy turned out to be essentially SigLIP-2's: per-member fine-tuned OOF was about 0.91 for DINOv3, 0.9435 for SigLIP-2, and about 0.91 for AIMv2, so blending the two weaker members did not lift the strongest. We then pushed a single SigLIP-2 with three low-overfit-risk levers (cosine head, class-balanced loss, optional hflip TTA). The cosine head plus balanced loss raised OOF to 0.9484 ± 0.0004; hflip TTA was a slight negative (about 1 image), so the deploy is identity-only.

5 Results

The deployed single SigLIP-2 reaches OOF 0.9484 ± 0.0004 (3-seed shared-fold mean; per-seed 0.9481 / 0.9481 / 0.9490). On the deploy seed, identity-only evaluation gives OOF 0.9490 versus 0.9481 with hflip TTA (TTA delta −0.0009, about 1 image), which is why the deploy is identity-only. The public leaderboard score is 0.93636, which ties the best score on the project (the fine-tuned 3-backbone ensemble) while being OOF-superior by +0.80 points and using one third of the fine-tuning cost (12 fine-tunes vs. 36) with roughly 6× tighter seed variance.

Table 2 and Figure 1 summarize how accuracy improved across the project. OOF and the leaderboard track together except for the reverted 512 px experiment.

Approach	OOF	Public LB
Frozen DINOv3 probe, 256 px (baseline)	0.8851	∼0.88
Frozen DINOv3 probe, 512 px (reverted)	0.8953	0.86363
Multi-view $K = 8$ frozen probe	0.9129	0.90000
LoRA DINOv3 (seed-42 4-fold)	0.9073	0.89090
Frozen cross-backbone ensemble	0.9314	0.91818
Fine-tuned LoRA ensemble (3 backbones)	0.9404	0.93636
Single SigLIP-2 (identity, deployed)	**0.9484**	**0.93636**

Table 2: Accuracy by approach across the project.

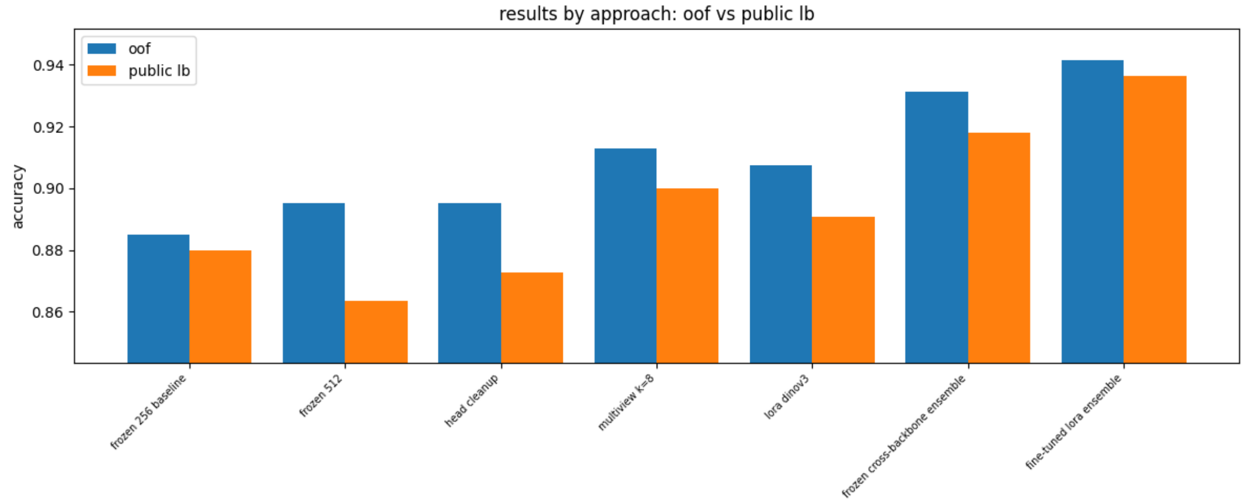

Figure 1: OOF and public leaderboard accuracy by approach (the bars run through the fine-tuned LoRA ensemble; the deployed single SigLIP-2 adds OOF 0.9484 / LB 0.93636 on top).

Figure 2 shows per-epoch training and validation loss and accuracy for the three fine-tuned backbones. SigLIP-2 and DINOv3 converge within about 10 to 15 epochs and plateau without collapsing; validation accuracy stays high while training accuracy approaches 1.0, indicating controlled (not catastrophic) overfitting thanks to LoRA, light augmentation, and early stopping.

The deployed submission (`outputs/submission_siglip2.csv`, copied to `outputs/submission.csv` for upload) scored 0.93636 on the public leaderboard. The repository README contains the leaderboard-position screenshot required by the assignment.

For error analysis, the OOF confusion matrix (Figure 3, for the comparable fine-tuned ensemble, whose accuracy the deployed single model matches and exceeds) is strongly diagonal: almost all mass is on the true class. The few brighter diagonal cells are simply the larger classes (up to 41 images); off-diagonal errors are sparse and scattered rather than concentrated in a confusable pair, which is consistent with a model that has learned all 100 classes rather than collapsing rare ones.

6 Discussion

Four things worked best. First, LoRA over full fine-tuning: with about 1000 images, adapting a small number of low-rank matrices in the last 4 attention blocks captured the task-specific signal without overfitting a 400M-parameter backbone. Second, fine-tuning a strong contrastive backbone: SigLIP-2's image-text pretraining transferred best and gained the most from fine-tuning (+4.4 points over its frozen probe), making it the single strongest model. Third, the cosine head with a class-balanced loss: on the imbalanced 4-to-41-per-class data, normalizing features and class weights and up-weighting rare classes contributed essentially all of the gain over a plain linear head (+0.80 points OOF), and it is a low-overfit-risk change. Fourth, trusting multi-seed OOF over the noisy public leaderboard: the roughly-110-image public slice could not separate the single model from the ensemble (both 0.93636) even though OOF separated them by +0.80 points, so we deployed the simpler, cheaper model.

The 512 px experiment is the clearest failure case: CV improved but the leaderboard regressed, a real CV-vs-test divergence most likely caused by a resolution shift between train and test. We held resolution fixed afterward. Test-time augmentation (hflip and scaled views) consistently hurt,

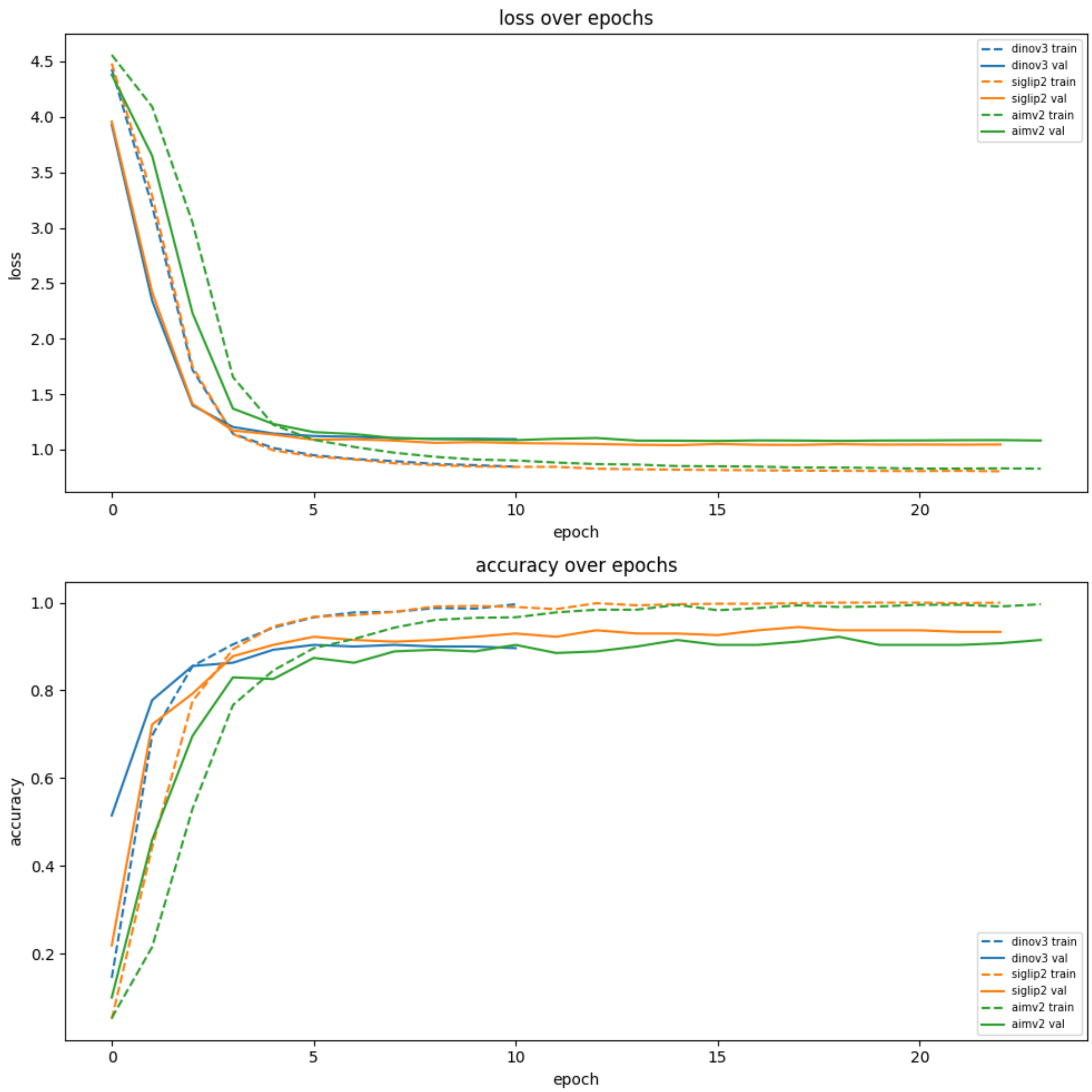

Figure 2: LoRA fine-tuning dynamics for the three backbones (train dashed, validation solid). The deployed model's dynamics are the SigLIP-2 curves.

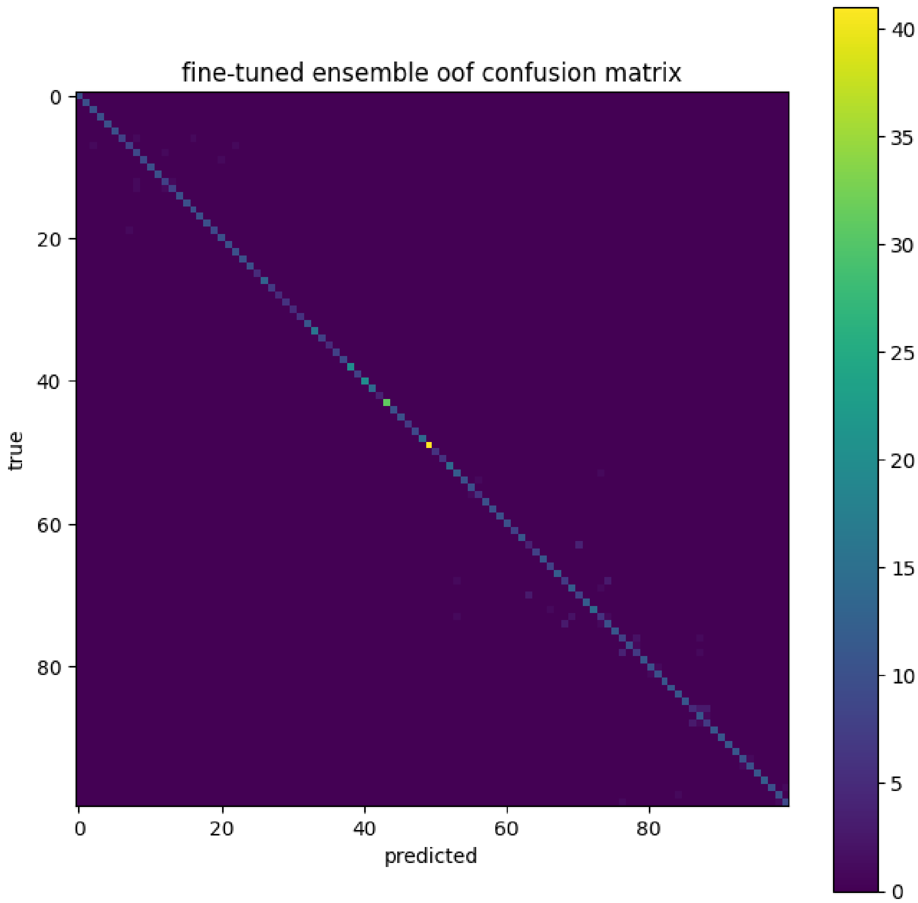

Figure 3: Out-of-fold confusion matrix (fine-tuned ensemble, comparable to the deployed single model). Predominantly diagonal; errors are sparse and not clustered.

both for the frozen probe at 512 px and for the fine-tuned model (about 1 image); averaging confident identity predictions with flipped views diluted them. Resolution ensembling and, after fine-tuning, backbone ensembling stopped helping: once SigLIP-2 was fine-tuned, the two weaker backbones could not lift it, so the ensemble's diversity benefit (which was real for the frozen models) evaporated. Overfitting is present but controlled: training accuracy approaches 1.0 while validation accuracy plateaus around 0.93 to 0.95 without collapsing, the expected behavior for LoRA with early stopping on a tiny dataset.

The main limitations are the tiny training set (1079 images) and the small, noisy public leaderboard (about 110 images), which limit how finely any single result can be trusted; we mitigate with multi-seed OOF but cannot eliminate the noise. We did not run a full per-lever ablation (cosine-only vs. balanced-only as separate training runs), so the combined gain is measured but not fully attributed; this is a cheap follow-up. Some bf16 attention operations are non-deterministic, so per-seed OOF drifts by about 0.2 to 0.3 points; we report a 3-seed mean ± std rather than a single number. Concrete next steps: the per-lever ablation; a careful capacity or resolution sweep (with a held-out leaderboard check, given the 512 px lesson); revisiting Sinkhorn test-time class balancing (the test set is approximately balanced); and more seeds for tighter variance estimates.

7 Reproducibility

A single `set_seed` function seeds Python `random`, NumPy, and PyTorch, sets cuDNN deterministic, and enables `torch.use_deterministic_algorithms` (with `warn_only`). Cross-validation uses a shared stratified 4-fold split (`StratifiedKFold` with `shuffle=True` and `random_state=seed`) so every comparison uses identical folds. Because some bf16 memory-efficient attention operations are non-deterministic, the deployed result is reported as a 3-seed (42, 43, 44) OOF mean ± std; per-seed accuracy can drift roughly 0.2 to 0.3 points, and may drift more across different torch/CUDA/GPU versions. The file `outputs/single_ft/metadata.json` records the exact git commit SHA and the torch/timm/scikit-learn/CUDA versions used.

The environment requires a CUDA 12.8 PyTorch build (RTX 50-series / Blackwell). It is verified with Python 3.14.0, torch 2.9.0+cu128, torchvision 0.24.0+cu128, timm 1.0.27, peft 0.19.1, scikit-learn 1.7.2, and numpy 2.3.2 on an RTX 5070 Ti (16 GB):

```
python -m venv .venv
.venv\Scripts\activate          # windows
# on linux/mac: source .venv/bin/activate
pip install -r requirements.txt
python -c "import torch; print(torch.cuda.is_available())"
```

The last command should print `True` on a GPU machine.

Run everything from the repository root. One command trains the deployed model and writes the submission:

```
python -m src.single_ft
```

This reads `config.yaml` (the `single_ft` block), runs the 3-seed shared-fold OOF, trains the seed-42 4-fold deploy ensemble, and writes `outputs/submission_siglip2.csv` (the deployed predictions, columns ID and Label, one row per test image; copy it to `outputs/submission.csv` to upload to Kaggle), `outputs/single_ft/single_ft_bundle.pkl` (the trained LoRA adapters and cosine heads), and `outputs/single_ft/metrics.json` and `metadata.json` (OOF scores, the TTA-vs-identity comparison, the decision outcome, and provenance).

Do not run the test suite from inside `data/` (a stray cache folder would be read as a class folder). The full suite (`pytest -q`) covers labels, data listing, submission validation, the LoRA building blocks (cosine head, class-balanced weights, TTA eval), the shared fold helpers, the single-model orchestrator, and the earlier pipelines. The trained model bundle is available at the Google Drive link in the repository README.

8 Team Contributions

This was a solo project.

9 References

1. R. Wightman, timm (PyTorch Image Models). Source of all pretrained backbone weights. `https://github.com/huggingface/pytorch-image-models`.
2. E. J. Hu et al., "LoRA: Low-Rank Adaptation of Large Language Models," 2021.
3. M. Tschannen et al., "SigLIP 2: Multilingual Vision-Language Encoders," 2025. Backbone `vit_so400m_patch14_siglip_gap_378.v2_webli`.
4. DINOv3 self-supervised vision transformers, Meta AI. Backbone `vit_large_patch16_dinov3.lvd1689m`.
5. AIMv2 autoregressive image models, Apple. Backbone `aimv2_large_patch14_336.apple_pt_dist`.
6. PyTorch (torch, torchvision), `https://pytorch.org`. PEFT (LoRA implementation), Hugging Face, `https://github.com/huggingface/peft`. scikit-learn, `https://scikit-learn.org`.
7. TorchVision pretrained models documentation, `https://pytorch.org/vision/stable/models.html`.
8. Kaggle competition: UCSC CSE 144 Spring 2026 Final Project, `https://www.kaggle.com/competitions/ucsc-cse-144-spring-2026-final-project`.